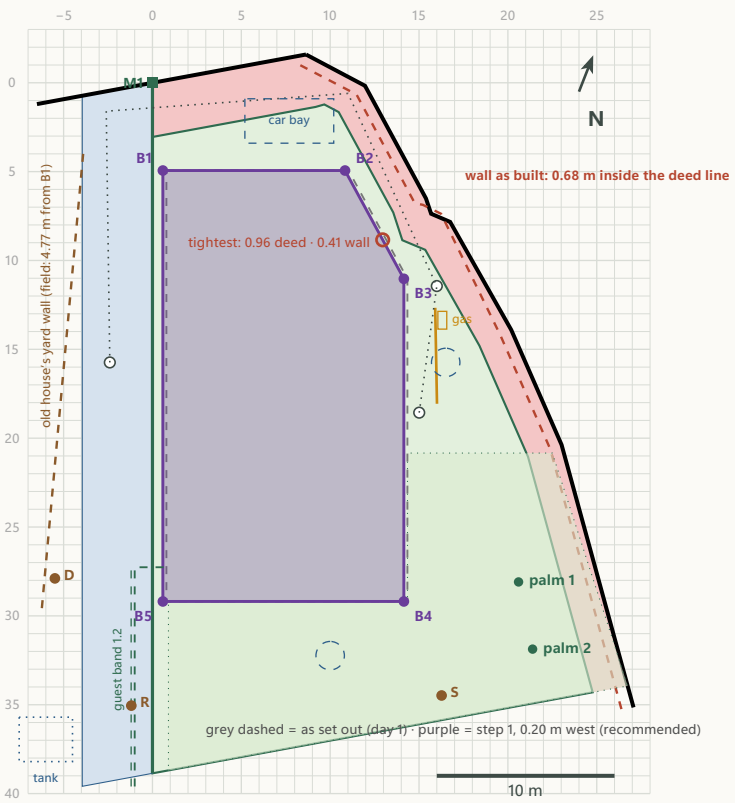


Step 1 · the whole house WEST, no rotation

Start: the house as set out on day 1 (B1 5.00 m from M1, the west outer face 0.79 m east of the M line). Options: the house unchanged, shifted 0.10 / 0.15 / 0.20 m west. Margins from the deed line and from the wall as built (0.68 m inside the deed line, the field value at B2). Every wall sampled at 5 cm.

Shift west (m)	**as set out**	**0.10**	**0.15**	**0.20** (rec.)
West outer face east of the M line = the strip to the driveway (m)	0.79	0.69	0.64	0.59
Tightest margin on any wall, deed (the cut wall at the lane's notch; ≥ 0.50 wanted)	**0.79**	**0.87**	**0.92**	**0.96**
Tightest margin, from the wall as built	**0.24**	**0.33**	**0.37**	**0.41**
B2's margin: deed / from the wall	1.08 / **0.40**	1.17 / **0.49**	1.21 / **0.53**	1.26 / **0.58**
B3 to the deed line / to the wall as built	3.67 / 2.99	3.76 / 3.07	3.80 / 3.12	3.84 / 3.16
B1 (and the west wall) to the driveway's west edge · to the old house's yard wall (4.77 as set out)	4.74 · 4.77	4.64 · 4.67	4.59 · 4.62	4.54 · 4.57
Driveway clear width (3.95 m): the cornice's edge (0.30 m) east of the M line; the band 0.08; the plinth flush; the guest door opens in	3.95 · **0.49**	3.95 · **0.39**	3.95 · **0.34**	3.95 · **0.28**
Guest path: the 1.2 m stone band on the driveway's edge (unchanged) + the strip to the wall = walking width	1.20 + 0.79 = 1.99	1.20 + 0.69 = 1.89	1.20 + 0.64 = 1.84	1.20 + 0.59 = 1.79
North strip: B1 to the north setback line · M1 → B1	2.00 · 5.00	1.98 · 4.99	1.97 · 4.98	1.96 · 4.97
Car bay: gap to the north wall (≥ 0.4) · east of the entrance path (≥ 0.1) · the reverse move's car side past B1	1.54 · 0.29 · 0.46	1.54 · 0.39 · 0.39	1.54 · 0.44 · 0.35	1.54 · 0.49 · 0.32
Palm 1 / palm 2 to the nearest wall	5.99 / 7.27	6.09 / 7.37	6.14 / 7.41	6.19 / 7.46
Gas cabinet to the wall as built (≥ 1.5) · tank to B5 (≥ 5) · the west drain to its chamber · soakaway E to the east wall	**1.79** · 8.39 · 3.31 · 2.15	**1.87** · 8.32 · 3.21 · 2.25	**1.92** · 8.29 · 3.16 · 2.30	**1.96** · 8.26 · 3.11 · 2.35
Garden lost (m²) · B5 to the garden's low wall	0.00 · 0.19	0.00 · 0.09	0.00 · 0.04	0.00 · −0.02
Conflicts	**none**	**none**	**none**	**none**

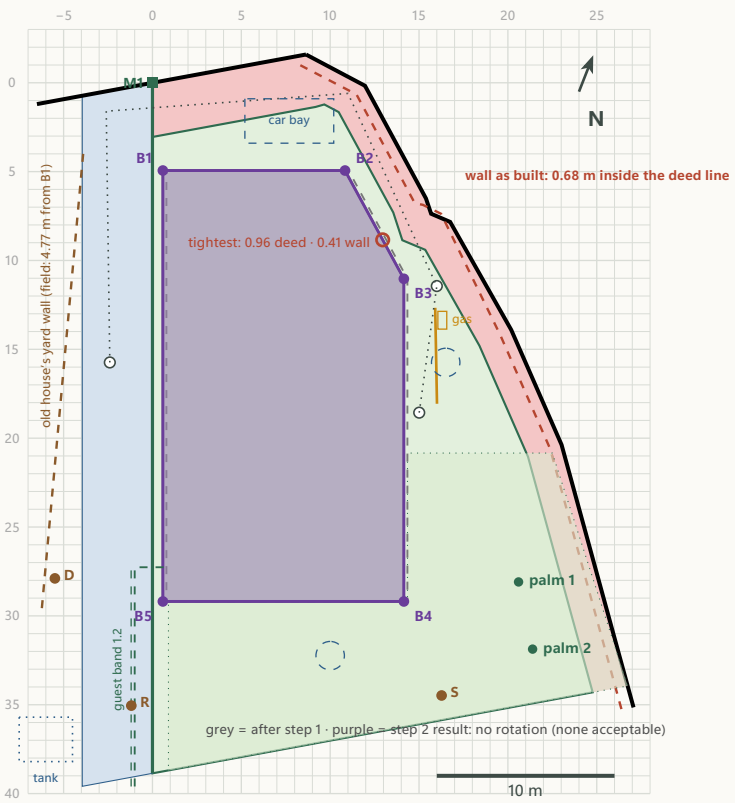
Recommendation · step 1: 0.20 m west

The shift is pure gain at the cut wall: every 0.10 m west adds 0.09 m of margin at the notch (the cut wall runs at 61° to the M line). From the wall as built the tightest margin goes 0.24 → **0.41 m** (the only option that reaches the owner's 0.40), B2 0.40 → 0.58 from the deed line 0.79 → 0.96. **The west side takes it:** the house stays 0.59 m east of the M line, the cornice 0.28 m clear of the driveway's edge, so the driveway keeps its 3.95 m at ground level and at cornice height; the guest band (1.2 m, on the driveway's edge) is untouched and the walking width wall-to-band-edge is 1.79 m; the north strip and the car bay are unchanged (the entrance path moves 0.20 m west with the house, the bay gains that clearance); the palms gain 0.20 m. **The costs:** the route across the driveway narrows from 4.77 to 4.57 m at B1 (still > the 3.95 m lane + the 0.585 strip; the water truck and the carport turn are at the south, unchanged); the strip beside the west wall is 0.59 m: keep it as the gravel / plinth drainage margin and the rescue-ladder stand, not as a path (the path is the band); the first-floor escape casements of bedrooms 2 and 3 on the west wall swing over the M line when open (they already do as set out: an open leaf reaches ≈ 0.3–0.5 m beyond the strip at +4.8 m above the ground, above any vehicle) — if the municipality objects, make them inward-opening or top-hung (the fire study allows either). The owner's rule B1 = 5.00 from M1 becomes 4.97 (B1 keeps its y).

Step 1 · set-out table of the recommended option (0.20 m west)

Corner	x east	y along M	Tape 1 (M1)	Tape 2 (D)	Tape 3 (R)	Tape 4 (S)	The block moves from the day-1 position
B1	0.59	4.94	M1 → **4.97**	D → **23.74**	R → **30.16**	S → **33.45**	−0.20 east, 0 south (0.20 m)
B2	10.84	4.94	M1 → **11.91**	D → **28.17**	R → **32.42**	S → **30.04**	−0.20 east, 0 south (0.20 m)
B3	14.15	11.03	M1 → **17.94**	D → **25.88**	R → **28.50**	S → **23.54**	−0.20 east, 0 south (0.20 m)
B4	14.15	29.19	M1 → **32.44**	D → **19.68**	R → **16.42**	S → **5.70**	−0.20 east, 0 south (0.20 m)
B5	0.59	29.19	M1 → **29.19**	D → **6.21**	R → **6.12**	S → **16.55**	−0.20 east, 0 south (0.20 m)

Sides B1–B2 10.25 · B2–B3 6.93 · B3–B4 18.16 · B4–B5 13.56 · B5–B1 24.25 · diagonals B1–B4 27.78 · B2–B5 26.33 · B1–B3 14.87 · perimeter 73.15. Every block moves 0.20 m straight west (the sides and the diagonals unchanged): tape the move from each block square to the string M1 → M4, then confirm with the diagonals.



Step 2 · a small rotation about B2, from the step-1 result

EAST · STOP rule: tightest margin from the wall as built < 0.40 m = not acceptable

Start: the step-1 result (0.20 m west). Options: B5 east by 0.10 / 0.20 / 0.30 m about B2 (B2 10.84, 4.94). The sides and the diagonals unchanged; the attached things (the yard, the gas, the chambers) turn with the house.

B5 east by (m)	**after step 1**	**0.10**	**0.20**	**0.30**
The house turns about B2 (°) · B1 moves south · M1 → B1	0 · 0 · 4.97	0.24 · 0.04 · 5.01	0.47 · 0.08 · 5.06	0.71 · 0.13 · 5.10
Tightest margin on any wall, deed (≥ 0.50 wanted)	**0.96**	**0.94**	**0.92**	**0.90**
Tightest margin, from the wall as built (STOP rule 0.40 three decimals, the 0.10 option is 5 mm short)	**0.414**	**0.395 not acceptable**	**0.377 not acceptable**	**0.358 not acceptable**
B2's margin: deed / from the wall (unchanged)	1.26 / **0.58**	1.26 / **0.58**	1.26 / **0.58**	1.26 / **0.58**
B3 to the deed line / to the wall as built	3.84 / 3.16	3.81 / 3.13	3.79 / 3.10	3.76 / 3.07
Route along the west wall: to the yard wall at B1 / y 15 / y 25 / B5 · to the driveway's edge at B5	4.57 / 5.48 / 6.39 / 6.77 · 4.54	4.57 / 5.53 / 6.48 / 6.88 · 4.64	4.58 / 5.57 / 6.56 / 6.98 · 4.74	4.58 / 5.61 / 6.64 / 7.09 · 4.84
Driveway clear width · the cornice's edge east of the M line (min along the wall)	3.95 · **0.28**	3.95 · **0.29**	3.95 · **0.29**	3.95 · **0.29**
Guest path: band 1.2 + strip at B1 / at B5 · the guest door moves east	1.20 + 0.59 / 0.59 · 0	1.20 + 0.59 / 0.69 · +0.09	1.20 + 0.59 / 0.79 · +0.18	1.20 + 0.59 / 0.89 · +0.28
Car bay: gap to the north wall · east of the entrance path · the reverse move past B1	1.54 · 0.49 · 0.32	1.56 · 0.51 · 0.35	1.58 · 0.53 · 0.38	1.61 · 0.55 · 0.41
Palm 1 / palm 2 · garden lost (m²)	6.19 / 7.46 · 0.00	6.09 / 7.37 · 0.21	6.00 / 7.28 · 0.41	5.90 / 7.20 · 0.99
Outdoor kitchen, screen, gas, grill turn with the house (spacings unchanged) · gas to the wall as built (≥ 1.5)	**1.96**	**1.92**	**1.88**	**1.83**
Tank to B5 (≥ 5) · sewer: yard chamber → east chamber · the west drain to its chamber · soakaway E	8.26 · 7.19 · 3.11 · 2.35	8.29 · 7.17 · 3.15 · 2.31	8.32 · 7.14 · 3.20 · 2.26	8.35 · 7.12 · 3.24 · 2.22
Conflicts	**none**	STOP rule: 0.40 < 0.40 from the wall as built	STOP rule: 0.38 < 0.40 from the wall as built	STOP rule: 0.36 < 0.40 from the wall as built

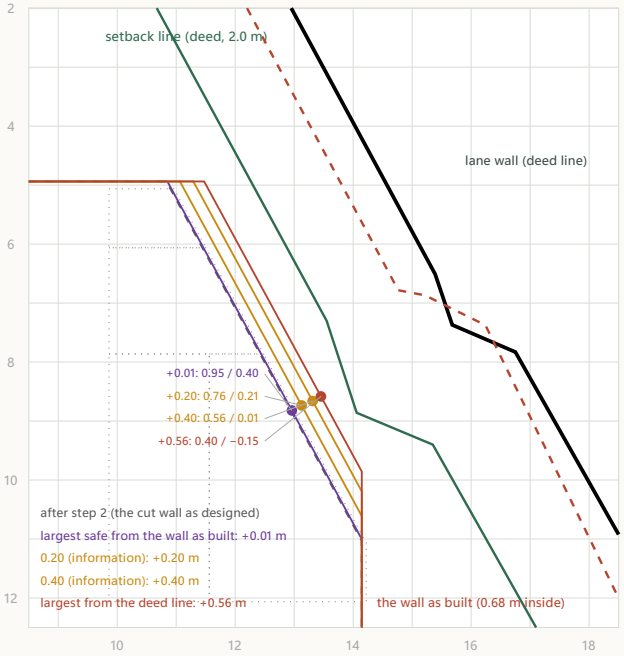
Recommendation · step 2: NO rotation

Every rotation turns the cut wall's far part towards the lane: the margin from the wall as built falls from 0.41 m (after step 1) to 0.395 / 0.377 / 0.358 m for 0.10 / 0.20 / 0.30 — all below the owner's 0.40 m STOP rule (the 0.10 m option by 0.005 m only). Against the deed line every option is fine (0.94 / 0.92 / 0.90 m, above the 0.50 spare), and nothing else conflicts (the palms, the garden, the guest path, the car bay, the yard, the gas, the tanks, the sewer all keep their rules). The rotation's gain (the route at the south: +0.10 … +0.30 m at B5, where the water truck and the carport turn are; the north strip at B1 +0.04 … +0.13 m) is not worth breaking the rule for. **So: no rotation; the step-1 position stands.** If the notch check (sheet row 18) finds the wall at the notch further off than the 0.68 m assumed here (the owner saw B2, not the notch, as the nearest point), the 0.10 m option becomes acceptable at a reading of ≥ 2.24 m from the present tightest point square to the wall (this study assumes 2.24 m there).

Step 2 · set-out table of the result (no rotation: the step-1 position stands)

Corner	x east	y along M	Tape 1 (M1)	Tape 2 (D)	Tape 3 (R)	Tape 4 (S)	The block moves from the day-1 position
B1	0.59	4.94	M1 → **4.97**	D → **23.74**	R → **30.16**	S → **33.45**	−0.20 east, 0 south (0.20 m)
B2	10.84	4.94	M1 → **11.91**	D → **28.17**	R → **32.42**	S → **30.04**	−0.20 east, 0 south (0.20 m)
B3	14.15	11.03	M1 → **17.94**	D → **25.88**	R → **28.50**	S → **23.54**	−0.20 east, 0 south (0.20 m)
B4	14.15	29.19	M1 → **32.44**	D → **19.68**	R → **16.42**	S → **5.70**	−0.20 east, 0 south (0.20 m)
B5	0.59	29.19	M1 → **29.19**	D → **6.21**	R → **6.12**	S → **16.55**	−0.20 east, 0 south (0.20 m)

Sides B1–B2 10.25 · B2–B3 6.93 · B3–B4 18.16 · B4–B5 13.56 · B5–B1 24.25 · diagonals B1–B4 27.78 · B2–B5 26.33 · B1–B3 14.87 · perimeter 73.15. No block moves in step 2: the position is step 1's (the table repeats step 1's values).



The cut wall enlarged. Grey dashed: after step 2. Purple: the largest move that keeps 0.40 m from the wall as built. Amber: 0.20 and 0.40 m, for information. Red: the largest move that keeps 0.40 m from the deed line. Each dot = the tightest point of that option with its margin deed / from the wall. The lane's notch (the deed vertex at 14.06, 8.86 and the wall vertex at 15.68, 7.37) governs: every move outward costs the same at the notch.

Step 3 · reducing the cut wall (the north-east slope), from the step-2 result

wall B2–B3 moved outwards, parallel · keeping ≥ 0.40 m from the wall as built · TO APPLY ONLY AFTER THE OFFICIAL SURVEY

Start: the step-2 result (no rotation: the house 0.20 m west of the day-1 position). The cut wall moves outwards by d; B2 slides east along the north wall (1.14 d) and B3 north along the east wall (2.09 d); the cut wall gets shorter. Each room along the cut gains ≈ d × its length along the cut.

Move outwards d (m)	**0.01** largest safe from the wall as built	**0.20** 0.20 (information)	**0.40** 0.40 (information)	**0.56** largest from the deed line
Tightest margin from the wall as built (≥ 0.40)	**0.40**	**0.21**	**0.01**	**−0.15**
Tightest margin from the deed line	**0.95**	**0.76**	**0.56**	**0.40**
B2 moves east · B3 moves north (m) · the cut wall's length (was 6.93)	0.02 · 0.03 · 6.90	0.23 · 0.42 · 6.46	0.46 · 0.84 · 5.98	0.64 · 1.17 · 5.60
Gross floor area gained per floor (m^2; the same on the roof)	**0.09**	**1.34**	**2.58**	**3.52**
Ground floor: the air tower (y 0–1 on the cut) · the maid bath (y 1–2.8) · the maid room (y 2.8–6)	0.02 · 0.03 · 0.05	0.23 · 0.41 · 0.73	0.46 · 0.82 · 1.46	0.64 · 1.15 · 2.04
Laundry · kitchen: they do not touch the cut wall (the maid room and the maid bath lie between): no gain unless the partitions are re-planned	0 · 0	0 · 0	0 · 0	0 · 0
First floor: the tower / HRV base (y 0–1) · bedroom 1 (y 1–6)	0.02 · 0.08	0.23 · 1.14	0.46 · 2.28	0.64 · 3.19
Bedroom 1 clear area (now 13.98 m^2; the hotel en-suite B4 needs 15.74 m^2 clear AND a rectangular bedroom)	14.06 · no: short by 1.68 m^2, and a wedge	15.12 · no: short by 0.62 m^2, and a wedge	16.26 · area OK, shape still a wedge	17.18 · area OK, shape still a wedge
The notch check (row 18, from the present tightest point square to the wall) must read at least this for the move to keep 0.40 from the wall as built	2.24	2.42	2.62	2.79

The columns and beams on the cut line (`docs/STRUCTURE_STUDY.md`: three columns at the cut's ends and at the maid bath / maid room partition, spans 3.2 · 3.6 m; the cut wall proposed as an RC wall W4): the corner column at B2 slides east along the north wall by the B2 move, the middle one outwards by d, the column at B3 north along the east wall by the B3 move; the north wall's last span grows by the B2 move (4.36 → 4.36 + 1.14 d), the east wall's first span shrinks by the B3 move (4.45 → 4.45 − 2.09 d); the cut-wall beams get shorter; the pads and the strip footing under W4 move with it. Nothing changes in the stiffness idea (the east box keeps its three RC walls). **The malqaf tower and the roof:** the air tower stands in the cut tip (x 8.95–10.59, y 0–1) with its east side on the cut line and the HRV base under it on the same line (`studies/roof.typ` Z2a): both grow outwards by d over 1.14 m of the cut (≈ 1.14 d m^2); the tower's louvred head keeps its size; the roof's cut edge, the main cornice and the attic's set-back follow the wall (the attic stays 1.40 m inside the cornice); the roof gains the same gross area; the sight-line checks of the roof study are not affected by a move of this size. **What it means:** with the wall as built taken 0.68 m inside the deed line all along, the cut wall after steps 1–2 keeps 0.41 m: **the largest safe move is 0.01 m — nothing worth building.** The 0.20 and 0.40 m moves (1.34 and 2.58 m^2 per floor, bedroom 1 +1.14 / +2.28 m^2) keep the deed line's margin but fall to 0.21 / 0.01 m from the wall as built; the largest move from the deed line (0.56 m, +3.52 m^2 per floor) would stand 0.15 m **inside** the setback counted from the wall. Bedroom 1's hotel en-suite B4 does not fit with any of them: it fails on shape (the room is a wedge) and, below ≈ 0.31 m, on area. **To apply only after the official survey** fixes the wall and the notch: the survey's reading at the notch (row 18) says which column applies.

Step 3 · set-out table of the largest safe move (+0.01 m) · information only, to apply only after the official survey

Corner	x east	y along M	Tape 1 (M1)	Tape 2 (D)	Tape 3 (R)	Tape 4 (S)	The block moves from the day-1 position
B1	0.59	4.94	M1 → **4.97**	D → **23.74**	R → **30.16**	S → **33.45**	−0.20 east, 0 south (0.20 m)
B2	10.85	4.94	M1 → **11.92**	D → **28.17**	R → **32.43**	S → **30.04**	−0.18 east, 0 south (0.18 m)
B3	14.15	11.00	M1 → **17.92**	D → **25.90**	R → **28.52**	S → **23.57**	−0.20 east, −0.03 south (0.20 m)
B4	14.15	29.19	M1 → **32.44**	D → **19.68**	R → **16.42**	S → **5.70**	−0.20 east, 0 south (0.20 m)
B5	0.59	29.19	M1 → **29.19**	D → **6.21**	R → **6.12**	S → **16.55**	−0.20 east, 0 south (0.20 m)

Sides B1–B2 10.27 · B2–B3 6.90 · B3–B4 18.19 · B4–B5 13.56 · B5–B1 24.25 · diagonals B1–B4 27.78 · B2–B5 26.33 · B1–B3 14.86 · perimeter 73.16. B2 slides 0.02 m east along the north wall, B3 0.03 m north along the east wall; B1, B4, B5 as step 2. The cut wall B2–B3 shortens to 6.90. **To apply only after the official survey.**

Summary · the three steps side by side and the recommended path

Measure	as set out (day 1)	Step 1 · 0.20 m west	Step 2 · no rotation	Step 3 · cut wall +0.01 m (after the survey)
Tightest margin on any wall: from the deed line / **from the wall as built**	**0.79** / **0.24**	**0.96** / **0.41**	**0.96** / **0.41**	**0.95** / **0.40**
B2's margin: deed / from the wall	1.08 / **0.40**	1.26 / **0.58**	1.26 / **0.58**	1.24 / **0.56**
B3 to the deed line / to the wall as built	3.67 / 2.99	3.84 / 3.16	3.84 / 3.16	3.83 / 3.15
Route beside the driveway: B1 to the yard wall · B5 to the yard wall · B5 to the driveway's edge	4.77 · 6.97 · 4.74	4.57 · 6.77 · 4.54	4.57 · 6.77 · 4.54	4.57 · 6.77 · 4.54
West wall east of the M line at B1 / B5 (the strip) · the cornice's edge clear of the driveway	0.79 / 0.79 · **0.49**	0.59 / 0.59 · **0.28**	0.59 / 0.59 · **0.28**	0.59 / 0.59 · **0.28**
Guest path: band 1.2 on the driveway + the strip = walking width at B1	1.99	1.79	1.79	1.79
M1 → B1 (the owner's 5.00 rule) · B1 to the north setback line	5.00 · 2.00	4.97 · 1.96	4.97 · 1.96	4.97 · 1.96
Car bay: gap to the north wall · east of the entrance path · the reverse move past B1	1.54 · 0.29 · 0.46	1.54 · 0.49 · 0.32	1.54 · 0.49 · 0.32	1.54 · 0.49 · 0.32
Palm 1 / palm 2 to the nearest wall	5.99 / 7.27	6.19 / 7.46	6.19 / 7.46	6.19 / 7.46
Gas cabinet to the wall as built (≥ 1.5) · tank to B5 (≥ 5) · soakaway E to the east wall	**1.79** · 8.39 · 2.15	**1.96** · 8.26 · 2.35	**1.96** · 8.26 · 2.35	**1.96** · 8.26 · 2.35
Gross floor area of the outline (m²) · gained per floor	318.77 · 0	318.77 · 0	318.77 · 0	318.86 · +0.09
Garden lost (m²)	0.00	0.00	0.00	0.00
What is gained	—	+0.18 m at the cut wall (from the wall); the palms +0.20	nothing (no option passes the STOP rule)	+0.09 m² per floor: nothing worth building
What is lost	—	the route at B1 4.77 → 4.57 the strip beside the west wall 0.79 → 0.59 B1 = 4.97 from M1	—	—
Blocks to move on site	—	all five, 0.20 m west	none	B2 and B3 only, after the survey

The recommended path

1 · Shift the house 0.20 m west (all five blocks, the house unchanged): the only step that buys margin at the cut wall — from the wall as built 0.24 → 0.41 m (the owner's 0.40 reached), from the deed line 0.79 → 0.96 m — at the price of a 0.59 m strip beside the west wall (the guest path stays the 1.2 m band on the driveway's edge; the cornice 0.28 m clear of the driveway) and a route of 4.57 m at B1. Tape table on page 1.

2 · No rotation: every option breaks the owner's STOP rule by 0.005–0.04 m from the wall as built (0.40 / 0.38 / 0.36) for a gain of 0.10–0.30 m at the south end only. The position after step 1 stands (page 2).

3 · The cut wall stays as designed until the official survey: the largest safe move from the wall as built is 0.01 m. If the survey (or the notch check, sheet row 18) shows the wall at the notch further off than the 0.68 m assumed here, page 3 says what 0.20 / 0.40 m would gain (1.34 / 2.58 m² per floor, bedroom 1 +1.14 / +2.28 m²; the hotel en-suite B4 still does not fit). Decision kept: no reduction of the cut wall until the official survey.

Before any block moves: the notch check (row 18: from the tightest point square to the lane wall; the deed says 2.92, the wall as built would give 2.24); then the owner's decision on step 1; then `design_gf.typ` gets the shift and the set-out sheet is re-issued (not done here).

How the two margins are read. The deed line is the plot's legal limit; the setback lines of the envelope (`base.typ`) are 2.0 m from the lane line and 3.0 m from the north line. The field value at B2 (2.40 m to the wall's inner face against 3.08 m to the deed line) says the wall stands 0.68 m inside the deed line; this study takes the same offset along the whole lane wall, which is the conservative reading (the owner saw B2 as the nearest point of the house to the wall, so the wall at the notch is probably further off than that). A setback counted from the wall as built is therefore the strict case; the deed-line margin the lenient one. The official survey replaces both.

What does not change in any step: the house itself (its sides, diagonals, rooms and the yard); the driveway's 3.95 m; the guest band; the car bay; the tanks, the septic and the sewer's falls; the palms' protection; the garden's low wall. **What changes:** the set-out tapes (page 1), the owner's rule B1 = 5.00 m (→ 4.97), the strip beside the west wall, the route's width at B1.

Placement in three steps · Study checks passed: the house unchanged in the design files (shift 0.91, 5.06 the outline asserted): study only · the calibration: the wall as built 0.68 m inside the deed line (0 … 1 m plausible); B1 → the yard wall 4.77 = the field value · step 1: every option keeps the house east of the M line with the cornice clear of the driveway (min 0.28 m), every corner inside the envelope, no conflict · step 1: the recommended shift (0.20 m) gives the largest margin from the wall as built (0.41) · step 2: B2 fixed, the sides and the diagonals unchanged in every option; the STOP rule applied (acceptable = tightest margin from the wall ≥ 0.40) · step 2: the recommendation is the largest acceptable rotation or "no rotation" (none acceptable) · step 3: the largest safe move keeps exactly 0.40 m from the wall as built (0.400) and the options never pass the deed line's 0.40 m · step 3: B2 and B3 slide along the north and east walls (B1, B4, B5 fixed; the cut stays parallel). Field values: the owner on site, day 1 (4 Nov 2026; docs/SITE_SETOUT.md § 7 rows 23–24). Sources: base.typ (the site v2), design_gf.typ (the house), site_plan.typ (the services), roof.typ, STRUCTURE_STUDY.md, bedroom1_bath.typ (B4), rotation_b2.typ (the calibration).